

Homework 1

PROBLEM 1 — SOLUTION

Heat into the gas and work done by the gas are positive. The gas is a closed system with boundary work only and negligible changes in bulk kinetic and potential energy. Here E is total internal energy and $e = E/m$. Use $R_u = 8.31446 \text{ J/(mol K)}$; bars denote molar quantities.

(a) (i) Isothermal: $T = \text{constant}$.

$$T_{2,\text{iso}} = T_1 = \boxed{300 \text{ K}}, \quad pV = nR_u T_1 = \text{constant}, \quad \frac{V_{2,\text{iso}}}{V_1} = \frac{p_1}{p_2} = \frac{1}{8}.$$

For a calorically perfect gas, $\Delta E = n\bar{c}_v(T_2 - T_1) = 0$. Thus the first law gives $0 = Q - W_{\text{by}}$, or $Q = W_{\text{by}}$. The reversible boundary work is

$$\begin{aligned} W_{\text{by,iso}} &= \int_{V_1}^{V_{2,\text{iso}}} p dV = nR_u T_1 \int_{V_1}^{V_{2,\text{iso}}} \frac{dV}{V} \\ &= nR_u T_1 \ln \frac{V_{2,\text{iso}}}{V_1} = -(1)(8.31446)(300) \ln 8 = \boxed{-5.187 \text{ kJ}}. \end{aligned}$$

(ii) Reversible adiabatic: $Q = 0$ and $S_{\text{gen}} = 0$, so $\Delta S = 0$.

$$\begin{aligned} pV^\gamma &= \text{constant}, & \frac{T_{2,\text{ad}}}{T_1} &= \left(\frac{p_2}{p_1}\right)^{(\gamma-1)/\gamma}, \\ T_{2,\text{ad}} &= 300(8)^{0.4/1.4} = \boxed{543.4 \text{ K}}. \end{aligned}$$

To obtain the molar heat capacity,

$$\begin{aligned} \bar{c}_p &= \bar{c}_v + R_u, & \gamma &= \frac{\bar{c}_p}{\bar{c}_v}, \\ (\gamma - 1)\bar{c}_v &= R_u, & \bar{c}_v &= \frac{R_u}{\gamma - 1} = 20.7862 \text{ J/(mol K)}. \end{aligned}$$

Apply the first law with $Q = 0$:

$$\Delta E = Q - W_{\text{by,ad}} = -W_{\text{by,ad}},$$

$$W_{\text{by,ad}} = -\Delta E = -m\Delta e = -n\bar{c}_v(T_{2,\text{ad}} - T_1) = \boxed{-5.060 \text{ kJ}}.$$

Unrounded intermediate values are used in the numerical calculations.

(b) Use the ideal-gas entropy relation and convert totals to mass-specific values:

$$\Delta S_{\text{sys}} = n \left(\bar{c}_p \ln \frac{T_2}{T_1} - R_u \ln \frac{p_2}{p_1} \right), \quad \bar{c}_p = \frac{\gamma R_u}{\gamma - 1},$$

$$m = n M_{\text{O}_2} = (1 \text{ mol})(0.03200 \text{ kg/mol}) = 0.03200 \text{ kg}, \quad \Delta e = \frac{\Delta E}{m}, \quad q = \frac{Q}{m}, \quad \Delta s_{\text{sys}} = \frac{\Delta S_{\text{sys}}}{m}.$$

Here $M_{\text{O}_2} = 32.00 \text{ g/mol}$ is the rounded molar mass used for the requested mass-specific results.

(i) **Isothermal:**

$$\Delta E = 0, \quad Q = W_{\text{by,iso}} = -5.187 \text{ kJ},$$

$$\Delta S_{\text{sys}} = n (\bar{c}_p \ln 1 - R_u \ln 8) = -n R_u \ln 8 = -17.29 \text{ J/K},$$

$$\boxed{\Delta e = 0}, \quad \boxed{q = -162.1 \text{ kJ/kg}}, \quad \boxed{\Delta s_{\text{sys}} = -0.5403 \text{ kJ/(kg K)}}.$$

(ii) **Reversible adiabatic:**

$$\Delta E = n \bar{c}_v (T_{2,\text{ad}} - T_1) = +5.060 \text{ kJ}, \quad Q = 0,$$

$$\Delta S_{\text{sys}} = n \left[\frac{\gamma R_u}{\gamma - 1} \frac{\gamma - 1}{\gamma} \ln 8 - R_u \ln 8 \right] = 0,$$

$$\boxed{\Delta e = +158.1 \text{ kJ/kg}}, \quad \boxed{q = 0}, \quad \boxed{\Delta s_{\text{sys}} = 0}.$$

(c) (i) **Isothermal:** The surroundings supply 5.187 kJ of work and receive 5.187 kJ of heat in a 300 K reservoir. For reversible heat transfer, the temperature difference tends to zero.

$$Q_{\text{surr}} = -Q = +5.187 \text{ kJ},$$

$$\Delta S_{\text{surr}} = \frac{Q_{\text{surr}}}{300 \text{ K}} = +17.29 \text{ J/K},$$

$$\Delta S_{\text{universe}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}} = -17.29 + 17.29 = \boxed{0}.$$

(ii) **Reversible adiabatic:** An ideal work source supplies 5.060 kJ; the insulated boundary permits no heat transfer.

$$\boxed{\Delta S_{\text{sys}} = \Delta S_{\text{surr}} = \Delta S_{\text{universe}} = 0}.$$

The ideal work source transfers energy without entropy. Both processes have zero entropy generation because they are reversible.

(d)

$$W_{\text{in}} = -W_{\text{by}}, \quad \boxed{W_{\text{in,iso}} = 5.187 \text{ kJ}}, \quad \boxed{W_{\text{in,ad}} = 5.060 \text{ kJ}},$$

$$V_{2,\text{iso}} = \frac{n R_u T_1}{p_2} = 3.118 \text{ L}, \quad V_{2,\text{ad}} = \frac{n R_u T_{2,\text{ad}}}{p_2} = 5.648 \text{ L}.$$

For these data, the adiabatic path requires 0.1268 kJ less work input. Its temperature rises, so it reaches 8 bar at a larger volume and stops sooner. The final pressures are equal, but the final states are different; comparing the paths at the same final volume would give a different work comparison.

PROBLEM 5 — SOLUTION

Let h and c denote the initially hot and cold blocks. Each rigid block has constant total heat capacity $C = 500 \text{ J/K}$. The combined system is isolated; kinetic and potential energy changes are negligible.

(a) For the two blocks together, $Q_{\text{ext}} = W_{\text{ext}} = 0$. Hence

$$\begin{aligned}\Delta E_h + \Delta E_c &= 0, \\ C(T_f - 500 \text{ K}) + C(T_f - 300 \text{ K}) &= 0, \\ T_f &= \boxed{400 \text{ K}}.\end{aligned}$$

The hot block loses 50.0 kJ and the cold block gains the same amount.

(b) The actual heat transfer is irreversible. Since entropy is a state property, use a reversible auxiliary path between each block's initial and final states:

$$\delta Q_{\text{rev}} = dE = C dT, \quad \Delta S = \int_{T_i}^{T_f} \frac{\delta Q_{\text{rev}}}{T} = C \ln \frac{T_f}{T_i}.$$

Thus

$$\begin{aligned}\Delta S_h &= (500 \text{ J/K}) \ln \frac{400}{500} = \boxed{-111.57 \text{ J/K}}, \\ \Delta S_c &= (500 \text{ J/K}) \ln \frac{400}{300} = \boxed{+143.84 \text{ J/K}}.\end{aligned}$$

The external surroundings are unchanged, so

$$\boxed{\Delta S_{\text{universe}} = \Delta S_h + \Delta S_c = 32.27 \text{ J/K} > 0.}$$

This is also the entropy generated within the isolated two-block system.

(c) The first law requires only that energy lost by one block equal energy gained by the other; it permits either sign of heat transfer. The second law selects the spontaneous direction. In a uniform-temperature description of each block, let $\delta Q > 0$ flow from the hot block to the cold block. Then

$$dS_{\text{universe}} = -\frac{\delta Q}{T_h} + \frac{\delta Q}{T_c} = \delta Q \left(\frac{1}{T_c} - \frac{1}{T_h} \right) > 0 \quad (T_h > T_c).$$

Heat flow in the reverse direction, with no other effect, would decrease the universe's entropy and violate the second law.

PROBLEM 7 — SOLUTION

(a) For a single-phase, fixed-composition substance, let h , s , and v be mass-specific properties. Since $h = h(T, p)$,

$$dh = \left(\frac{\partial h}{\partial T} \right)_p dT + \left(\frac{\partial h}{\partial p} \right)_T dp = c_p dT + \left(\frac{\partial h}{\partial p} \right)_T dp.$$

Using $dh = T ds + v dp$ and the mass-specific Gibbs energy $g = h - Ts$,

$$dg = dh - T ds - s dT = (T ds + v dp) - T ds - s dT = -s dT + v dp.$$

$$\left(\frac{\partial g}{\partial T} \right)_p = -s, \quad \left(\frac{\partial g}{\partial p} \right)_T = v.$$

The mixed derivatives of the state function g are equal, so

$$\frac{\partial}{\partial p} \left(\frac{\partial g}{\partial T} \right)_p = \frac{\partial}{\partial T} \left(\frac{\partial g}{\partial p} \right)_T \implies \left(\frac{\partial s}{\partial p} \right)_T = - \left(\frac{\partial v}{\partial T} \right)_p.$$

Therefore, at fixed T ,

$$\left(\frac{\partial h}{\partial p} \right)_T = T \left(\frac{\partial s}{\partial p} \right)_T + v = v - T \left(\frac{\partial v}{\partial T} \right)_p,$$

and substitution gives

$$\boxed{dh = c_p dT + \left[v - T \left(\frac{\partial v}{\partial T} \right)_p \right] dp.}$$

(b) With the mass-specific gas constant R , $Z = pv/(RT)$ gives

$$v = \frac{ZRT}{p}, \quad \left(\frac{\partial v}{\partial T} \right)_p = \frac{R}{p} \left[Z + T \left(\frac{\partial Z}{\partial T} \right)_p \right].$$

Thus,

$$\begin{aligned} v - T \left(\frac{\partial v}{\partial T} \right)_p &= \frac{ZRT}{p} - \frac{RT}{p} \left[Z + T \left(\frac{\partial Z}{\partial T} \right)_p \right] \\ &= \frac{ZRT}{p} - \frac{ZRT}{p} - \frac{RT^2}{p} \left(\frac{\partial Z}{\partial T} \right)_p = -\frac{RT^2}{p} \left(\frac{\partial Z}{\partial T} \right)_p. \end{aligned}$$

Hence

$$\boxed{dh = c_p dT - \frac{RT^2}{p} \left(\frac{\partial Z}{\partial T} \right)_p dp.}$$

For an ideal gas, $Z = 1$, so the pressure term vanishes.

(c) The supplied B is molar; use $\bar{v}$, $\bar{h}$, and $R_u = 8.314462618 \text{ J mol}^{-1}\text{K}^{-1}$. Also,

$$\mathcal{M} = 0.0280134 \text{ kg mol}^{-1}, \quad \Delta p = (200 - 1) \times 10^5 = 1.99 \times 10^7 \text{ Pa}.$$

Within the supplied pressure-truncated model,

$$\begin{aligned} Z &= 1 + \frac{B(T)p}{R_u T}, & \bar{v} &= \frac{Z R_u T}{p} = \frac{R_u T}{p} + B(T), \\ \left(\frac{\partial \bar{v}}{\partial T} \right)_p &= \frac{R_u}{p} + B'(T), & B'(T) &\equiv \frac{dB}{dT}. \end{aligned}$$

Since $dT = 0$, the molar form of part (a) gives

$$\begin{aligned} d\bar{h} &= \left[\frac{R_u T}{p} + B - T \left(\frac{R_u}{p} + B' \right) \right] dp = (B - T B') dp, \\ \Delta \bar{h} &= \int_{p_1}^{p_2} (B - T B') dp = (B - T B') \Delta p. \end{aligned}$$

$$\boxed{\Delta h = \frac{\Delta \bar{h}}{\mathcal{M}} = \frac{(B - T B') \Delta p}{\mathcal{M}}}.$$

(i) **Rigid spheres.** With $N_A = 6.02214076 \times 10^{23} \text{ mol}^{-1}$,

$$\begin{aligned} B_{\text{RS}} &= \frac{2\pi}{3} N_A \sigma^3 = \frac{2\pi}{3} (6.02214076 \times 10^{23}) (3.681 \times 10^{-10})^3 \\ &= 6.2908158 \times 10^{-5} \text{ m}^3 \text{ mol}^{-1}, \quad B'_{\text{RS}} = 0, \\ \Delta \bar{h}_{\text{RS}} &= (6.2908158 \times 10^{-5}) (1.99 \times 10^7) = 1251.872 \text{ J mol}^{-1}, \\ \Delta h_{\text{RS}} &= \frac{1251.872}{0.0280134} \text{ J kg}^{-1} = \boxed{+44.69 \text{ kJ kg}^{-1}}. \end{aligned}$$

(ii) **Attraction-inclusive model.**

$$\begin{aligned} B - T B' &= [-75.7 - 150(1.01)] \times 10^{-6} = -227.2 \times 10^{-6} \text{ m}^3 \text{ mol}^{-1}, \\ \Delta \bar{h} &= (-227.2 \times 10^{-6}) (1.99 \times 10^7) = -4521.280 \text{ J mol}^{-1}, \\ \Delta h &= \frac{-4521.280}{0.0280134} \text{ J kg}^{-1} = \boxed{-161.4 \text{ kJ kg}^{-1}}. \end{aligned}$$

At fixed T , the same model gives

$$\Delta \bar{e} = \Delta \bar{h} - \Delta(p\bar{v}) = (B - T B') \Delta p - B \Delta p = -T B' \Delta p.$$

Rigid spheres have $B' = 0$: internal energy is unchanged, and the positive enthalpy change comes from $p\bar{v}$. Attractions introduce a negative internal-energy contribution; here $B - T B' < 0$, reversing the enthalpy trend.

At 200 bar, however,

$$Z_{\text{RS}} = 2.0088, \quad Z_{\text{attraction}} = -0.21395.$$

The rigid-sphere correction is not small, and the negative Z in model (ii) is unphysical. These are formal results of the assigned truncation; model (ii) fails before the final pressure is reached.

PROBLEM 8 — SOLUTION

Assume steady, isentropic flow of ideal nitrogen, with no heat transfer or shaft work, negligible elevation change, and $V_1 \simeq 0$. Here V denotes flow speed, h is mass-specific enthalpy, and a bar denotes a molar quantity.

$$T_1 = 1200 \text{ K}, \quad \frac{p_2}{p_1} = \frac{1}{50}, \quad M = 0.0280134 \text{ kg/mol}, \quad R_u = 8.314462618 \text{ J/(mol K)}.$$

(a) For the calorically perfect gas (CPG), $\gamma = 1.40$ and $\Delta s = 0$:

$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1}\right)^{(\gamma-1)/\gamma} \implies T_{2,\text{CPG}} = 1200 \left(\frac{1}{50}\right)^{0.4/1.4} = 392.429 \text{ K}.$$

(b) For the thermally perfect gas (TPG), $\bar{c}_p(T) = 27.0 + 0.0040T$ in J/(mol K). Since $\Delta \bar{s} = 0$,

$$\begin{aligned} 0 &= \int_{T_1}^{T_2} \frac{\bar{c}_p(T)}{T} dT - R_u \ln \frac{p_2}{p_1}, \\ \int_{1200}^{T_2} \left(\frac{27.0}{T} + 0.0040\right) dT &= R_u \ln \frac{1}{50}, \\ 27.0 \ln \frac{T_2}{1200} + 0.0040(T_2 - 1200) &= 8.314462618 \ln \frac{1}{50}. \end{aligned}$$

Solving with MATLAB `fzero` on $300 < T_2 < 500$ K gives

$$T_{2,\text{TPG}} = 404.728 \text{ K}.$$

The left-hand side increases strictly for $T_2 > 0$, since its derivative is $27.0/T_2 + 0.0040 > 0$. Thus the root is unique and satisfies $0 < T_2 < T_1$.

(c) For both models, the positive static enthalpy drop is

$$h_1 - h_2 = \frac{1}{M} \int_{T_2}^{T_1} \bar{c}_p(T) dT, \quad h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2} \implies V_2 \simeq \sqrt{2(h_1 - h_2)}.$$

CPG:

$$\begin{aligned} \bar{c}_p &= \frac{\gamma R_u}{\gamma - 1} = 29.100619 \text{ J/(mol K)}, \\ h_1 - h_2 &= \frac{\bar{c}_p}{M} (T_1 - T_2) = \frac{29.100619}{0.0280134} (1200 - 392.429) \simeq 838.913 \text{ kJ/kg}, \\ V_{2,\text{CPG}} &= \sqrt{2(838.913 \times 10^3)} \simeq 1295.31 \text{ m/s}. \end{aligned}$$

TPG:

$$\begin{aligned} h_1 - h_2 &= \frac{1}{M} \int_{T_2}^{T_1} (27.0 + 0.0040T) dT \\ &= \frac{27.0(T_1 - T_2) + 0.0020(T_1^2 - T_2^2)}{M} \\ &= \frac{27.0(1200 - 404.728) + 0.0020(1200^2 - 404.728^2)}{0.0280134} \\ &\simeq 857.616 \text{ kJ/kg}, \\ V_{2,\text{TPG}} &= \sqrt{2(857.616 \times 10^3)} \simeq 1309.67 \text{ m/s}. \end{aligned}$$

In each speed calculation, the enthalpy drop is converted to J/kg; $1 \text{ J/kg} = 1 \text{ m}^2/\text{s}^2$.

(d) Both models assume an ideal gas and the same steady, adiabatic, isentropic nozzle flow. CPG uses constant heat capacities; TPG uses temperature-dependent heat capacities. The specified TPG law gives a higher exit temperature but a larger mean heat capacity over its expansion. Its enthalpy drop and exit speed are therefore slightly larger. Neither model includes intermolecular real-gas effects.

PROBLEM 11 — SOLUTION

Subscripts 1 and 2 denote methane and argon, respectively; subscript 0 denotes an initial state. Treat both gases as ideal and calorically perfect, with the prescribed degrees of freedom. The outer vessel is rigid and insulated, with no shaft work or changes in endpoint bulk kinetic or potential energy. Neglect the partition's volume and heat capacity. Here E and S are totals, V is volume, and $R_u = 8.3144626 \text{ J/(mol K)}$.

(a) The partition is fixed: $V_{i,f} = V_{i,0}$. For the combined system, $Q_{\text{ext}} = W_{\text{ext}} = 0$. First calculate the amounts and the total heat capacities C_i of the gas samples:

$$\begin{aligned} n_1 &= \frac{m_1}{M_1} = \frac{1}{0.0160425} = 62.334424 \text{ mol}, & n_2 &= \frac{1}{0.039948} = 25.032542 \text{ mol}, \\ \bar{c}_{v,i} &= \frac{f_i}{2} R_u, & C_i &= n_i \bar{c}_{v,i} = m_i c_{v,i}, \\ C_1 &= n_1(3R_u) = 1554.831716 \text{ J/K}, & C_2 &= n_2 \left(\frac{3}{2} R_u \right) = 312.198205 \text{ J/K}. \end{aligned}$$

The first law gives

$$\begin{aligned} \Delta E_1 + \Delta E_2 &= 0, \\ C_1(T_{f,a} - 300) + C_2(T_{f,a} - 600) &= 0, \\ T_{f,a} &= \frac{C_1(300) + C_2(600)}{C_1 + C_2} = \boxed{350.164949 \text{ K}}. \end{aligned}$$

For each gas, the entropy change is

$$\Delta S_i = C_i \ln \frac{T_{i,f}}{T_{i,0}} + n_i R_u \ln \frac{V_{i,f}}{V_{i,0}}.$$

Since the compartment volumes are unchanged, $\ln(V_{i,f}/V_{i,0}) = 0$:

$$\begin{aligned} \Delta S_{1,a} &= 1554.831716 \ln \frac{350.164949}{300} = \boxed{+240.411 \text{ J/K}}, \\ \Delta S_{2,a} &= 312.198205 \ln \frac{350.164949}{600} = \boxed{-168.127 \text{ J/K}}, \\ \Delta S_a &= \Delta S_{1,a} + \Delta S_{2,a} = \boxed{72.2843 \text{ J/K}} > 0. \end{aligned}$$

The system is isolated, so $S_{\text{gen},a} = \Delta S_a$. Heat transfer across a finite temperature difference is irreversible. The final temperature is closer to 300 K because methane has the larger total heat capacity.

The fixed partition can support unequal final pressures:

$$p_{i,a} = p_{i,0} \frac{T_{f,a}}{T_{i,0}}, \quad p_{1,a} = 1.16721650 \text{ bar}, \quad p_{2,a} = 0.58360825 \text{ bar}.$$

(b) Return to the original initial states. The ideal-gas internal energies depend only on temperature, so removing the partition adds no energy-of-mixing term:

$$C_1(T_{f,b} - 300) + C_2(T_{f,b} - 600) = 0 \implies \boxed{T_{f,b} = T_{f,a} = 350.164949 \text{ K}}.$$

From the initial ideal-gas states,

$$\begin{aligned} V_{i,0} &= \frac{n_i R_u T_{i,0}}{p_{i,0}}, & p_{i,0} &= 10^5 \text{ Pa}, \\ V_{1,0} &= 1.554831716 \text{ m}^3, & V_{2,0} &= 1.248792821 \text{ m}^3, \\ V &= V_{1,0} + V_{2,0} = 2.803624536 \text{ m}^3. \end{aligned}$$

The final mixture pressure is

$$p_f = \frac{(n_1 + n_2) R_u T_{f,b}}{V} = 90726.52 \text{ Pa} = \boxed{0.907265 \text{ bar}}.$$

Each species now occupies the entire vessel, so

$$\Delta S_{i,b} = C_i \ln \frac{T_{f,b}}{T_{i,0}} + n_i R_u \ln \frac{V}{V_{i,0}} = \Delta S_{i,a} + n_i R_u \ln \frac{V}{V_{i,0}}.$$

Therefore,

$$\begin{aligned} \Delta S_{1,b} &= 240.410961 + (62.334424)(8.3144626) \ln \frac{2.803624536}{1.554831716} = \boxed{545.959 \text{ J/K}}, \\ \Delta S_{2,b} &= -168.126641 + (25.032542)(8.3144626) \ln \frac{2.803624536}{1.248792821} = \boxed{+0.197252 \text{ J/K}}, \\ \Delta S_b &= \Delta S_{1,b} + \Delta S_{2,b} = \boxed{546.156 \text{ J/K}} = S_{\text{gen},b} > 0. \end{aligned}$$

Unrounded values are retained: argon's cooling and volume terms nearly cancel, but its net entropy change is positive.

(c) The additional entropy is

$$\Delta S_{\text{extra}} = \Delta S_b - \Delta S_a \simeq 546.156 - 72.2843 = \boxed{473.872 \text{ J/K}}.$$

In (b), each gas can spread through the entire vessel, whereas in (a) it stays in its original compartment. I think this additional volume contribution is related to ideal-gas mixing. I have not yet worked out how to connect it to the usual mole-fraction expression for mixing entropy.